12. Completeness of L^p and pointwise convergence

REAL VARIABLES I

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In Note 24 we proved that $\|\cdot\|_p$ is a norm on $L^p(X,\mu)$ for $p \in [1,+\infty)$ and a quasinorm for $p \in (0,1)$. In Note 1 we saw that the space $C^0([a,b])$ of continuous functions and the space of Riemann-integrable functions are not complete with respect to the L^1 distance: some Cauchy sequences of such functions have no limit within the space, in the same way that a Cauchy sequence of rational numbers may converge to an irrational number. Completeness is the property that allows us to construct an object as the limit of approximations, for instance as the sum of a series, and to know that the limit belongs to the space under study. These notes prove that $L^p(X,\mu)$ is complete for every exponent $p \in (0,+\infty)$. This theorem is known as the **Riesz–Fischer theorem**.

The proof works with the quantity $\|f\|_p^p = \int_X |f|^p \, \mathrm{d}\mu$ rather than with $\|f\|_p$. Because $\|f\|_p^p$ is the integral of the pointwise quantity $|f|^p$, the monotone convergence theorem applies to it directly, and a single quasi-triangle inequality $\|f+g\|_p^p \le K_p(\|f\|_p^p + \|g\|_p^p)$ covers all exponents $p \in (0,+\infty)$ simultaneously. A lemma reduces completeness to the convergence of Cauchy sequences whose consecutive differences satisfy $\|f_{n+1} - f_n\|_p^p \le C^{-n}$, and for such sequences we construct the limit pointwise, as the sum of a telescoping series.

The construction produces a limit that is simultaneously an L^p limit and an almost-everywhere limit. In the second part of these notes we compare these two modes of convergence. We prove that L^p convergence with summable errors implies almost-everywhere convergence, deduce that every L^p-convergent sequence has an almost-everywhere convergent subsequence, and construct counterexamples showing that neither mode of convergence implies the other in general.

After studying these notes, you will be able to do the following.

- State the quasi-triangle inequality for $\|\cdot\|_p^p$ with constant $K_p = \max\{1, 2^{p-1}\}$ and iterate it to bound the p-th power of a finite sum.
- Reduce the convergence of a Cauchy sequence in $L^p(X,\mu)$ to the convergence of a subsequence with geometrically decaying consecutive differences.
- Prove the Riesz–Fischer theorem: $L^p(X,\mu)$ is complete for every $p \in (0,+\infty)$.
- Prove that L^p convergence with $\sum_n \|f_n - f\|_p^p < +\infty$ implies almost-everywhere convergence, and that every L^p-convergent sequence has an almost-everywhere convergent subsequence.
- Construct a sequence that converges in L^p but converges at no point, and sequences that converge everywhere but have no limit in L^p.
- Apply the dominated convergence theorem to deduce L^p convergence from almost-everywhere convergence.

Throughout these notes, $(X, \mathcal{A}, \mu)$ denotes a measure space, $p \in (0,+\infty)$ is a fixed exponent, and functions are measurable functions $f\colon X \to \mathbb{R}$. On subsets of $\mathbb{R}$, the symbol $\mathcal{L}$ denotes Lebesgue measure.

12.1 Cauchy sequences in L^p

We first adapt the notions of convergence and of Cauchy sequences from Note 1 to the space $L^p(X,\mu)$. Because the function $t \mapsto t^{1/p}$ is an increasing bijection of $[0,+\infty)$ onto itself, a quantity $\|h\|_p$ tends to 0 if and only if $\|h\|_p^p$ tends to 0. We may therefore formulate both notions with $\|\cdot\|_p^p$.

Definition 1 (Convergence and Cauchy sequences in L^p). A sequence $(f_n)_{n\in\mathbb{N}}$ in $L^p(X,\mu)$ **converges in** L^p to $f \in L^p(X,\mu)$ if

$$\lim_{n\to\infty} \|f_n - f\|_p^p = \lim_{n\to\infty} \int_X |f_n - f|^p \, \mathrm{d}\mu = 0.$$

The sequence $(f_n)_{n\in\mathbb{N}}$ is a **Cauchy sequence in** L^p if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$\|f_n - f_m\|_p^p < \varepsilon \qquad \text{for all } n, m \geq N.$$

The space $L^p(X,\mu)$ is **complete** if every Cauchy sequence in L^p converges in L^p to an element of $L^p(X,\mu)$.

For $p \in [1,+\infty)$, these notions are convergence and the Cauchy property with respect to the metric $d_p(f,g) = \|f-g\|_p$; for $p \in (0,1)$, they are the same notions with respect to the metric $d_p(f,g) = \|f-g\|_p^p$ (Note 24). In both cases Definition 1 agrees with the definition of a complete metric space in Note 1.

12.1.1 A quasi-triangle inequality for the p-th power of the norm

All estimates in these notes rest on one inequality for the p-th power $\|\cdot\|_p^p$. Its constant depends on whether p lies below or above 1:

$$K_p := \max\{1, 2^{p-1}\} = \begin{cases} 1, & 0 < p \leq 1, \\ 2^{p-1}, & p > 1. \end{cases} \tag{1}$$

Proposition 2 (Quasi-triangle inequality for $\|\cdot\|_p^p$). For all real numbers $a, b \geq 0$,

$$(a+b)^p \leq K_p(a^p + b^p). \tag{2}$$

Consequently, for all $f, g \in L^p(X,\mu)$,

$$\|f+g\|_p^p \leq K_p(\|f\|_p^p + \|g\|_p^p). \tag{3}$$

Proof of Proposition 2. Step 1: The numerical inequality.

For $p \in (0,1]$, the inequality $(a+b)^p \leq a^p + b^p$ is the subadditivity of $t \mapsto t^p$ proved in Step 1 of the proof that $\|\cdot\|_p$ is a quasinorm (Note 24); for $p = 1$ it is an equality. For $p > 1$, the function $t \mapsto t^p$ is convex on $(0,+\infty)$ (Note 24), and the argument of Step 2 of the same proof, applied with the exponent $r = p$, gives $(a+b)^p \leq 2^{p-1}(a^p + b^p)$ for all $a, b \geq 0$. In both cases (2) holds.

Step 2: Integration.

At every point $x \in X$, the triangle inequality in $\mathbb{R}$, the monotonicity of $t \mapsto t^p$ on $[0,+\infty)$, and (2) with $a = |f(x)|$ and $b = |g(x)|$ give

$$|f(x)+g(x)|^p \leq \big(|f(x)| + |g(x)|\big)^p \leq K_p(|f(x)|^p + |g(x)|^p).$$

Integrating over X yields (3). □

For $p \in (0,1]$, the inequality (3) is the triangle inequality for $\|\cdot\|_p^p$ from Note 24. For $p > 1$, Minkowski's inequality gives the sharper bound $\|f+g\|_p \leq \|f\|_p + \|g\|_p$, but the weaker inequality (3) suffices for every argument below and allows a single proof for all exponents.

The inequality (3) also shows that L^p limits are unique. If $f_n \to f$ and $f_n \to g$ in L^p, then

$$\|f-g\|_p^p \leq K_p(\|f-f_n\|_p^p + \|f_n-g\|_p^p) \longrightarrow 0 \qquad \text{as } n \to \infty,$$

so $\|f-g\|_p = 0$ and $f = g$ in $L^p(X,\mu)$.
The completeness proof applies (2) to sums with many terms. Applying the two-term inequality repeatedly multiplies the constant K_p at each step, and the next lemma records the resulting weights.

Lemma 3 (Iterated quasi-triangle inequality). For every $N \in \mathbb{N}$ and all real numbers $a_1, \dots, a_N \geq 0$,

$$\left(\sum_{n=1}^{N} a_n\right)^p \leq \sum_{n=1}^{N} K_p^n \, a_n^p. \tag{4}$$

Proof of Lemma 3. We argue by induction on N. For $N = 1$, the inequality reads $a_1^p \leq K_p a_1^p$, which holds because $K_p \geq 1$. Suppose that (4) holds for sums of $N-1$ terms, where $N \geq 2$. Applying (2) with $a = a_1$ and $b = \sum_{n=2}^{N} a_n$, and then the induction hypothesis to the $N-1$ numbers $a_2, \dots, a_N$, gives

$$\left(\sum_{n=1}^{N} a_n\right)^p \leq K_p a_1^p + K_p \left(\sum_{n=2}^{N} a_n\right)^p \leq K_p a_1^p + K_p \sum_{n=2}^{N} K_p^{n-1} a_n^p = \sum_{n=1}^{N} K_p^n \, a_n^p.$$

This proves (4). □

For $p \in (0,1]$, all weights in (4) equal 1. For $p > 1$, the weights K_p^n grow geometrically in n. A series $\sum_n a_n$ can therefore be controlled through (4) only when the terms a_n^p decay faster than K_p^{-n}; this requirement determines the decay rate in the reduction lemma below.

12.1.2 Reduction to rapidly Cauchy sequences

A Cauchy sequence guarantees that $\|f_n - f_m\|_p^p$ is eventually small, but it gives no rate at which this quantity decreases. To build the limit as the sum of the telescoping series $f_1 + \sum_n (f_{n+1} - f_n)$, we need consecutive differences that decay at a prescribed geometric rate. The next lemma shows that it is enough to prove convergence for sequences with this rate: every Cauchy sequence has a subsequence with geometrically decaying differences, and a Cauchy sequence converges as soon as one of its subsequences converges.

Lemma 4 (Reduction to rapidly Cauchy sequences). Suppose that $C > 1$ and that every sequence $(g_k)_{k\in\mathbb{N}}$ in $L^p(X,\mu)$ satisfying

$$\|g_{k+1} - g_k\|_p^p \leq C^{-k} \qquad \text{for all } k \in \mathbb{N} \tag{5}$$

converges in L^p to an element of $L^p(X,\mu)$. Then every Cauchy sequence in $L^p(X,\mu)$ converges in L^p, so $L^p(X,\mu)$ is complete.

Proof of Lemma 4. Suppose that $(f_n)_{n\in\mathbb{N}}$ is a Cauchy sequence in $L^p(X,\mu)$.

Step 1: Extraction of a rapidly Cauchy subsequence.

For each $k \in \mathbb{N}$, the Cauchy property with $\varepsilon = C^{-k}$ provides an index N_k such that $\|f_n - f_m\|_p^p < C^{-k}$ for all $n, m \geq N_k$. We choose indices $n_1 < n_2 < n_3 < \cdots$ with $n_k \geq N_k$ for every k. Because both n_k and n_{k+1} are at least N_k, the subsequence $g_k \coloneqq f_{n_k}$ satisfies

$$\|g_{k+1} - g_k\|_p^p = \|f_{n_{k+1}} - f_{n_k}\|_p^p < C^{-k} \qquad \text{for all } k \in \mathbb{N}.$$

By hypothesis, $(g_k)_{k\in\mathbb{N}}$ converges in L^p to some $f \in L^p(X,\mu)$.

Step 2: Convergence of the whole sequence.

Fix $\varepsilon > 0$ and choose $N \in \mathbb{N}$ such that $\|f_n - f_m\|_p^p < \varepsilon$ for all $n, m \geq N$. Because $n_k \to \infty$ and $\|f_{n_k} - f\|_p^p \to 0$, there exists k with $n_k \geq N$ and $\|f_{n_k} - f\|_p^p < \varepsilon$. For every $n \geq N$, the quasi-triangle inequality (3) gives

$$\|f_n - f\|_p^p \leq K_p(\|f_n - f_{n_k}\|_p^p + \|f_{n_k} - f\|_p^p) < 2K_p\varepsilon.$$

Because $\varepsilon > 0$ was arbitrary, $\|f_n - f\|_p^p \to 0$. This proves that $f_n \to f$ in L^p. □

12.2 The Riesz–Fischer theorem

Theorem 5 (Riesz–Fischer theorem). For every $p \in (0, +\infty)$, the space $L^p(X, \mu)$ is complete: every Cauchy sequence in $L^p(X, \mu)$ converges in L^p to an element of $L^p(X, \mu)$.

By Lemma 4, it suffices to prove that every sequence satisfying (5) converges in L^p, for one fixed choice of $C > 1$. We choose $C \coloneqq 2K_p$, so that the geometric weights K_p^n of Lemma 3 are dominated by the decay C^{-n}: the products $K_p^n C^{-n} = 2^{-n}$ are summable. The proof then follows the telescoping series

$$f_n = f_1 + \sum_{k=1}^{n-1} (f_{k+1} - f_k)$$

at each point $x \in X$. Step 1 shows that the series of absolute values $\sum_k |f_{k+1}(x) - f_k(x)|$ is finite for almost every x; Step 2 defines the limit f pointwise; Step 3 estimates the tails of the series in L^p; Step 4 concludes.

Proof of Theorem 5. Set $C \coloneqq 2K_p$ and suppose that $(f_n)_{n\in\mathbb{N}}$ is a sequence in $L^p(X, \mu)$ satisfying

$$\|f_{n+1} - f_n\|_p^p \leq C^{-n} \qquad \text{for all } n \in \mathbb{N}.$$

We fix a representative of each f_n, which is a measurable function $f_n \colon X \to \mathbb{R}$, and define the measurable functions $h_n \coloneqq |f_{n+1} - f_n| \colon X \to [0, +\infty)$.

Step 1: The series of absolute differences is finite almost everywhere.

We define $G \colon X \to [0, +\infty]$ by $G(x) \coloneqq \sum_{n=1}^{+\infty} h_n(x)$. For every $N \in \mathbb{N}$, Lemma 3 applied pointwise to $h_1(x), \dots, h_N(x)$ and integrated over X gives

$$\int_X \left(\sum_{n=1}^{N} h_n \right)^p \mathrm{d}\mu \leq \sum_{n=1}^{N} K_p^n \int_X h_n^p \, \mathrm{d}\mu \leq \sum_{n=1}^{N} K_p^n C^{-n} = \sum_{n=1}^{N} 2^{-n} \leq 1.$$

As $N \to \infty$, the partial sums $\sum_{n=1}^{N} h_n$ increase pointwise to G, so their p-th powers increase pointwise to G^p. The monotone convergence theorem yields $\int_X G^p \, \mathrm{d}\mu \leq 1$. A non-negative function with finite integral is finite almost everywhere (Note 21), so the set $E \coloneqq \{x \in X \colon G(x) = +\infty\}$ satisfies $\mu(E) = 0$.

Step 2: Definition of the pointwise limit.

For $x \in X \setminus E$, the series $\sum_{k=1}^{+\infty} (f_{k+1}(x) - f_k(x))$ converges absolutely, so the partial sums

$$f_n(x) = f_1(x) + \sum_{k=1}^{n-1} (f_{k+1}(x) - f_k(x))$$

converge to a real number as $n \to \infty$. We define $f(x) \coloneqq \lim_{n\to\infty} f_n(x)$ for $x \in X \setminus E$ and $f(x) \coloneqq 0$ for $x \in E$. The function f is measurable as the pointwise limit of the measurable functions $f_n \mathbb{1}_{X\setminus E}$, and $f_n \to f$ almost everywhere.

Step 3: Tail estimate in L^p.

For $n \in \mathbb{N}$, we define the tail $T_n \colon X \to [0, +\infty]$ by $T_n(x) \coloneqq \sum_{k=n}^{+\infty} h_k(x)$. For $x \in X \setminus E$, passing to the limit $m \to \infty$ in $|f_m(x) - f_n(x)| \le \sum_{k=n}^{m-1} h_k(x)$ gives

$$|f(x) - f_n(x)| \le T_n(x). \tag{6}$$

For every $M \in \mathbb{N}$, Lemma 3 applied to the $M+1$ numbers $h_n(x), h_{n+1}(x), \dots, h_{n+M}(x)$ assigns the weight K_p^{j+1} to $h_{n+j}(x)$. Integrating over X gives

$$\int_X \left(\sum_{j=0}^{M} h_{n+j} \right)^p \mathrm{d}\mu \le \sum_{j=0}^{M} K_p^{j+1} C^{-(n+j)} = K_p C^{-n} \sum_{j=0}^{M} 2^{-j} \le 2K_p C^{-n}.$$

As $M \to \infty$, the partial sums increase pointwise to T_n, and the monotone convergence theorem yields $\int_X T_n^p \, \mathrm{d}\mu \le 2K_p C^{-n}$. Because $\mu(E) = 0$, the bound (6) holds almost everywhere, and therefore

$$\int_X |f - f_n|^p \, \mathrm{d}\mu \le \int_X T_n^p \, \mathrm{d}\mu \le 2K_p C^{-n}. \tag{7}$$

Step 4: Conclusion.

By (7), the function $f - f_n$ belongs to $\mathcal{L}^p(X, \mu)$. Because $\mathcal{L}^p(X, \mu)$ is a vector space (Note 24), $f = f_n + (f - f_n) \in \mathcal{L}^p(X, \mu)$. Since $C > 1$, the bound (7) gives $\|f_n - f\|_p^p \le 2K_p C^{-n} \to 0$, so $f_n \to f$ in L^p. Lemma 4 now shows that every Cauchy sequence in $L^p(X, \mu)$ converges in L^p. □

Corollary 6 (L^p as a complete space). For $p \in [1, +\infty)$, the normed space $(L^p(X, \mu), \|\cdot\|_p)$ is complete; a complete normed space is called a **Banach space**. For $p \in (0, 1)$, the space $L^p(X, \mu)$ with the metric $d_p(f, g) = \|f - g\|_p^p$ is a complete metric space.

In particular, $L^1([a, b], \mathcal{L})$ is complete. The Cauchy sequences of continuous functions constructed in Note 1, which have no limit in $C^0([a, b])$, converge in $L^1([a, b], \mathcal{L})$ to a limit that is not represented by any continuous function.

The proof of Theorem 5 established more than L^p convergence. In Step 2, the rapidly Cauchy sequence (f_n) also converged to f almost everywhere. Combined with the subsequence extraction of Lemma 4, this observation shows that every Cauchy sequence in L^p has a subsequence that converges almost everywhere to its L^p limit. The next section examines the relation between L^p convergence and pointwise convergence directly.

12.3 L^p convergence and pointwise convergence

A sequence of functions can approach a limit f in two different senses. Almost-everywhere convergence requires $f_n(x) \to f(x)$ at almost every individual point x. Convergence in L^p requires the averaged quantity $\int_X |f_n - f|^p \, \mathrm{d}\mu$ to tend to 0, and it allows $f_n(x)$ to differ substantially from $f(x)$ on sets whose measure tends to 0. In this section we prove that rapid L^p convergence implies almost-everywhere convergence, and we construct examples showing that neither mode of convergence implies the other in general.

12.3.1 Rapid L^p convergence implies almost-everywhere convergence

Theorem 7 (Rapid L^p convergence implies almost-everywhere convergence). Suppose that $f_n, f \in L^p(X, \mu)$ satisfy

$$\sum_{n=1}^{+\infty} \|f_n - f\|_p^p < +\infty. \tag{8}$$

Then $f_n \to f$ almost everywhere.

The proof turns the summability of the integrals $\int_X |f_n - f|^p \, \mathrm{d}\mu$ into the summability of the values $|f_n(x) - f(x)|^p$ at almost every point; the terms of a convergent series tend to 0.

Proof of Theorem 7. We define $S \colon X \to [0, +\infty]$ by $S(x) \coloneqq \sum_{n=1}^{+\infty} |f_n(x) - f(x)|^p$. The partial sums of this series are non-negative measurable functions that increase pointwise to S. The monotone convergence theorem and the additivity of the integral for non-negative functions give

$$\int_X S \, \mathrm{d}\mu = \sum_{n=1}^{+\infty} \int_X |f_n - f|^p \, \mathrm{d}\mu = \sum_{n=1}^{+\infty} \|f_n - f\|_p^p < +\infty.$$

A non-negative function with finite integral is finite almost everywhere (Note 21), so $S(x) < +\infty$ for almost every $x \in X$. At every such point, the terms of the convergent series $\sum_n |f_n(x) - f(x)|^p$ tend to 0, which gives $f_n(x) \to f(x)$. □

Corollary 8 (Almost-everywhere convergent subsequences). If $f_n \to f$ in L^p, then there exist indices $n_1 < n_2 < \cdots$ such that $f_{n_k} \to f$ almost everywhere as $k \to \infty$.

Proof of Corollary 8. Because $\|f_n - f\|_p^p \to 0$, we may choose indices $n_1 < n_2 < \cdots$ such that $\|f_{n_k} - f\|_p^p < 2^{-k}$ for every $k \in \mathbb{N}$. The subsequence $(f_{n_k})_{k \in \mathbb{N}}$ satisfies $\sum_k \|f_{n_k} - f\|_p^p < \sum_k 2^{-k} = 1$, and Theorem 7 gives $f_{n_k} \to f$ almost everywhere. □

Corollary 9 (Agreement of L^p and almost-everywhere limits). If $f_n \to f$ in L^p and $f_n \to g$ almost everywhere, then $f = g$ almost everywhere.

Proof of Corollary 9. By Corollary 8, a subsequence (f_{n_k}) converges to f outside a null set N_1. The same subsequence converges to g outside the null set N_2 on which (f_n) fails to converge to g. At every $x \notin N_1 \cup N_2$, the real sequence $(f_{n_k}(x))$ has the two limits $f(x)$ and $g(x)$, so $f(x) = g(x)$. Because $\mu(N_1 \cup N_2) = 0$, this proves that $f = g$ almost everywhere. □

Corollary 9 shows that the two modes of convergence never produce conflicting limits. Whenever a sequence converges in both senses, the L^p limit and the almost-everywhere limit are the same element of $L^p(X, \mu)$.

12.3.2 L^p convergence without pointwise convergence

Corollary 8 asserts almost-everywhere convergence only along a subsequence. The following example shows that passing to a subsequence cannot be avoided: an L^p-convergent sequence may fail to converge at every point.

Example 10 (The typewriter sequence). On $[0,1)$ with Lebesgue measure, every positive integer n can be written uniquely as $n = 2^k + j$ with integers $k \geq 0$ and $0 \leq j < 2^k$. We define

$$f_n \coloneqq \mathbb{1}_{I_n}, \qquad I_n \coloneqq \left[\frac{j}{2^k}, \frac{j+1}{2^k}\right), \qquad n = 2^k + j.$$

For each fixed k, the indices $n = 2^k, 2^k+1, \ldots, 2^{k+1}-1$ form the k-th *generation*, and the intervals I_n of that generation are the 2^k dyadic intervals of length 2^{-k}, which partition $[0,1)$ from left to right (Figure 1).

Convergence in L^p. For $n = 2^k + j$, we have $\|f_n\|_p^p = \mathcal{L}(I_n) = 2^{-k}$. Because $n < 2^{k+1}$, this gives $\|f_n - 0\|_p^p = 2^{-k} < 2/n$, so $f_n \to 0$ in L^p.

Divergence at every point. Fix $x \in [0,1)$. In each generation k, the point x belongs to exactly one of the intervals I_n, so $f_n(x) = 1$ for infinitely many n. In each generation $k \geq 1$, the point x lies outside the remaining $2^k - 1 \geq 1$ intervals, so $f_n(x) = 0$ for infinitely many n. Hence the sequence $(f_n(x))_{n\in\mathbb{N}}$ takes both values 0 and 1 infinitely often and does not converge, for every $x \in [0,1)$.

A rapidly convergent subsequence. The subsequence of first intervals of each generation, $n_k \coloneqq 2^k$, gives $f_{n_k} = \mathbb{1}_{[0,2^{-k})}$ and $\sum_k \|f_{n_k}\|_p^p = \sum_k 2^{-k} < +\infty$. In agreement with Theorem 7, $f_{n_k}(x) \to 0$ for every $x \in (0,1)$, that is, almost everywhere. For the full sequence, $\sum_n \|f_n\|_p^p = \sum_{k\geq 0} 2^k \cdot 2^{-k} = +\infty$, so Theorem 7 does not apply.

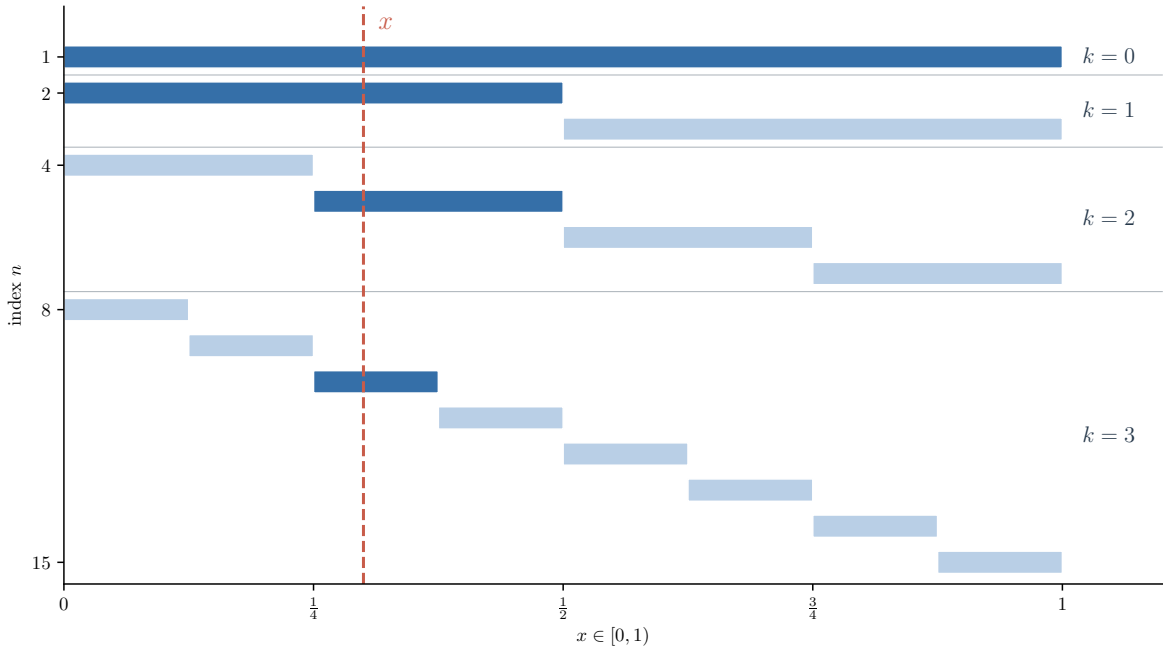


Figure 1: The intervals I_n of the typewriter sequence for $n = 1, \ldots, 15$, grouped by generation k. Each generation sweeps across $[0,1)$ with intervals of length 2^{-k}. A fixed point x (dashed line) lies in exactly one interval of every generation (dark bars), so $f_n(x) = 1$ infinitely often.

Figure description: A chart with the index n on the vertical axis and the interval from 0 to 1 on the horizontal axis. Each row n shows a horizontal bar over the interval I sub n. Rows are grouped into generations k equals 0 to 3, with one bar of full length, then two halves, four quarters, and eight eighths. A vertical dashed line at a fixed point x crosses one dark bar in every generation.

12.3.3 Pointwise convergence without L^p convergence

Conversely, a sequence may converge at every point without converging in L^p. The two mechanisms identified in Note 22, escape of mass to infinity and concentration of mass, both produce such sequences once the heights are adjusted to the exponent p.

Example 11 (Escape to infinity and concentration in L^p). On $\mathbb{R}$ with Lebesgue measure, the translates $f_n := \mathbb{1}_{[n,n+1)}$ satisfy $f_n(x) \to 0$ for every $x \in \mathbb{R}$, because $x \notin [n, n+1)$ once $n > x$. However, $\|f_n\|_p^p = \mathcal{L}([n, n+1)) = 1$ for every $n \in \mathbb{N}$.
On $(0, 1]$ with Lebesgue measure, the functions

$$g_n := 2^{n/p}\, \mathbb{1}_{[2^{-n}, 2^{1-n})}, \qquad n \in \mathbb{N},$$

satisfy $g_n(x) \to 0$ for every $x \in (0, 1]$, because $x \in [2^{-n}, 2^{1-n})$ for at most one index n. However, $\|g_n\|_p^p = 2^n\, \mathcal{L}([2^{-n}, 2^{1-n})) = 2^n \cdot 2^{-n} = 1$ for every $n \in \mathbb{N}$.
Neither sequence converges in L^p to any element of L^p. Indeed, suppose that $f_n \to h$ in L^p for some $h \in L^p(\mathbb{R})$. Because $f_n \to 0$ everywhere, Corollary 9 gives $h = 0$ almost everywhere, and then $\|f_n\|_p^p = \|f_n - h\|_p^p \to 0$, which contradicts $\|f_n\|_p^p = 1$. The same argument applies to (g_n). By Theorem 5, neither sequence is a Cauchy sequence in L^p.

The second sequence in Example 11 lives on the finite measure space $(0, 1]$. Finiteness of the measure therefore does not make almost-everywhere convergence imply L^p convergence.

12.3.4 Domination restores L^p convergence

In Note 22 the dominated convergence theorem excluded both escape to infinity and concentration by requiring a single integrable bound for the whole sequence. Applied to the functions $|f_n - f|^p$, the same theorem converts almost-everywhere convergence into L^p convergence under a domination hypothesis in L^p.

Proposition 12 (Dominated convergence in L^p). Suppose that measurable functions $f_n \colon X \to \mathbb{R}$ converge almost everywhere to a measurable function $f \colon X \to \mathbb{R}$, and that there exists $g \in L^p(X, \mu)$ with $g \geq 0$ such that

$$|f_n(x)| \leq g(x) \qquad \text{for almost every } x \in X \text{ and every } n \in \mathbb{N}.$$

Then $f_n, f \in L^p(X, \mu)$ and $f_n \to f$ in L^p.

Proof of Proposition 12. Because $|f_n|^p \leq g^p$ almost everywhere and $\int_X g^p \, \mathrm{d}\mu < +\infty$, each f_n belongs to $L^p(X, \mu)$. Passing to the limit in $|f_n(x)| \leq g(x)$ gives $|f| \leq g$ almost everywhere, so $f \in L^p(X, \mu)$. The functions $u_n := |f_n - f|^p$ converge to 0 almost everywhere and satisfy

$$u_n \leq (|f_n| + |f|)^p \leq (2g)^p = 2^p g^p \qquad \text{almost everywhere,}$$

where $2^p g^p$ is integrable. The dominated convergence theorem (Note 22) gives $\int_X u_n \, \mathrm{d}\mu \to 0$, that is, $\|f_n - f\|_p^p \to 0$. □

In Example 11, the smallest candidate dominating functions $\sup_n |f_n| = \mathbb{1}_{[1,+\infty)}$ and $\sup_n |g_n|$ do not belong to L^p: the first has infinite integral of its p-th power, and the p-th power of the second has the non-integrable singularity of $1/x$ at 0, as in Note 22. On a finite measure space, every sequence that is uniformly bounded by a constant M is dominated by the constant function M, which belongs to $L^p(X, \mu)$ because $\int_X M^p \, \mathrm{d}\mu = M^p \mu(X) < +\infty$.

Remark (Summary of the relations between the two modes of convergence). For $p \in (0, +\infty)$ and functions $f_n, f \in L^p(X, \mu)$:

- if $\sum_n \|f_n - f\|_p^p < +\infty$, then $f_n \to f$ almost everywhere (Theorem 7);
- if $f_n \to f$ in L^p, then a subsequence converges to f almost everywhere (Corollary 8), but the whole sequence may converge at no point (Example 10);
- if $f_n \to f$ almost everywhere, then (f_n) need not converge in L^p, even on a finite measure space (Example 11), but it does converge in L^p to f when $|f_n| \leq g$ for a single $g \in L^p(X, \mu)$ (Proposition 12);
- if a sequence converges both in L^p and almost everywhere, the two limits agree almost everywhere (Corollary 9).